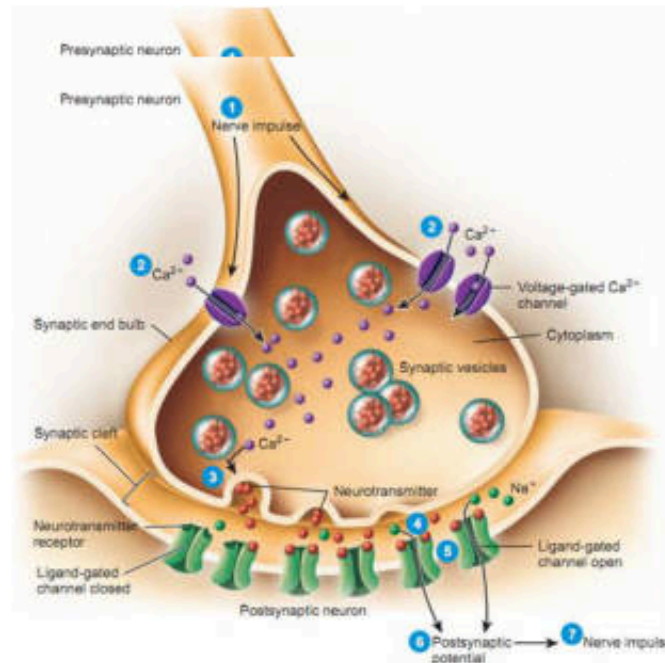




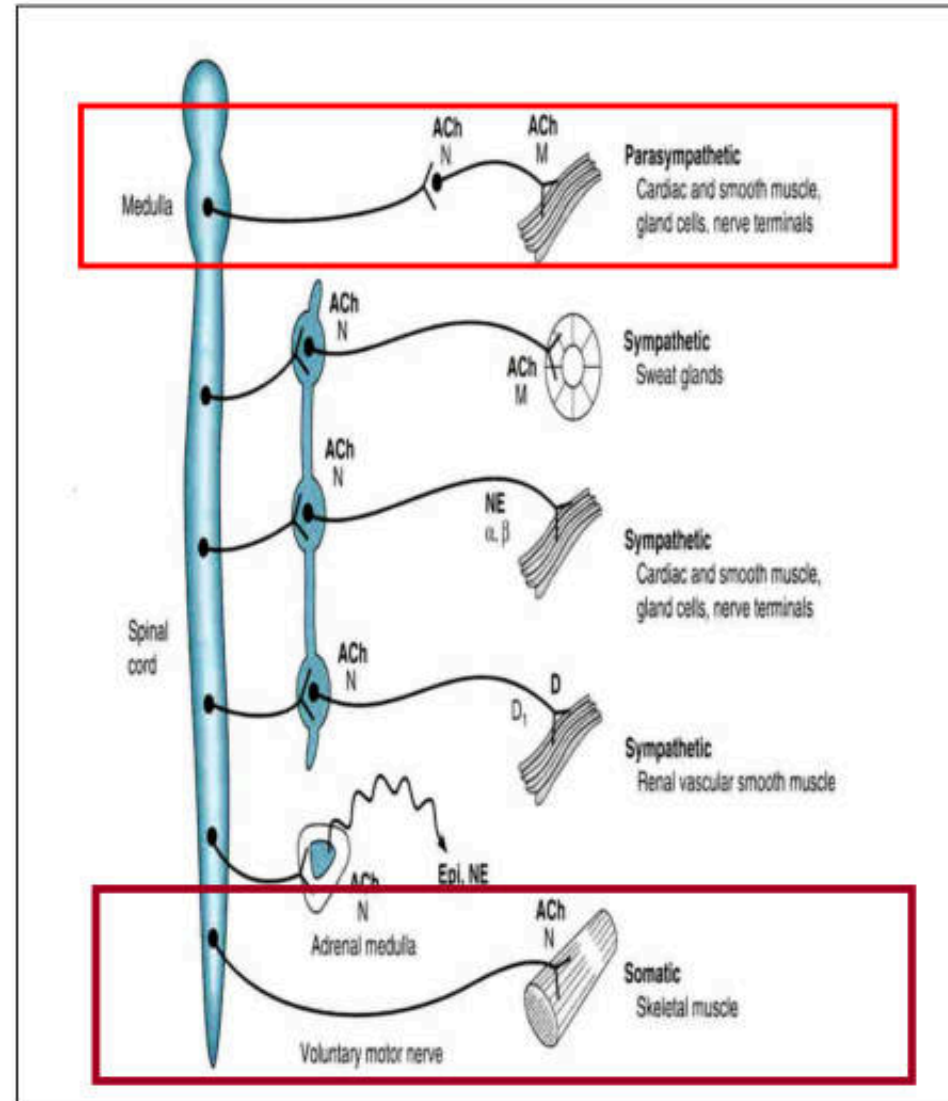
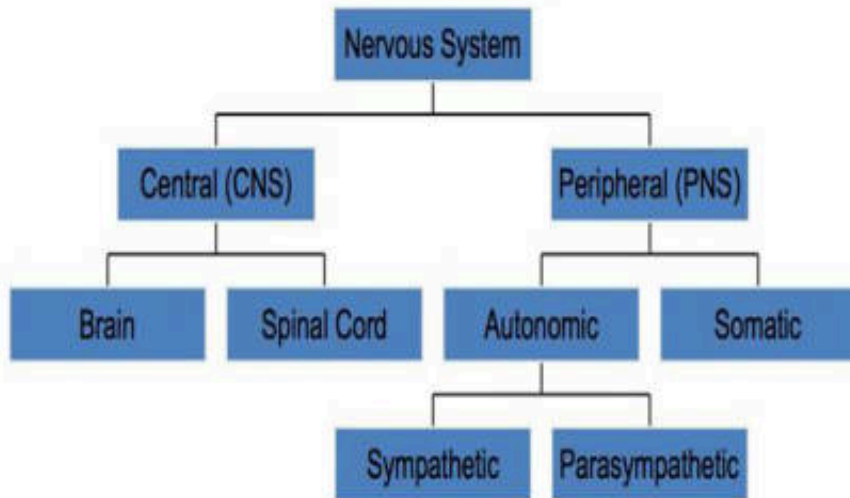
Pharmacology of Drugs Used in Parasympathetic Nervous System



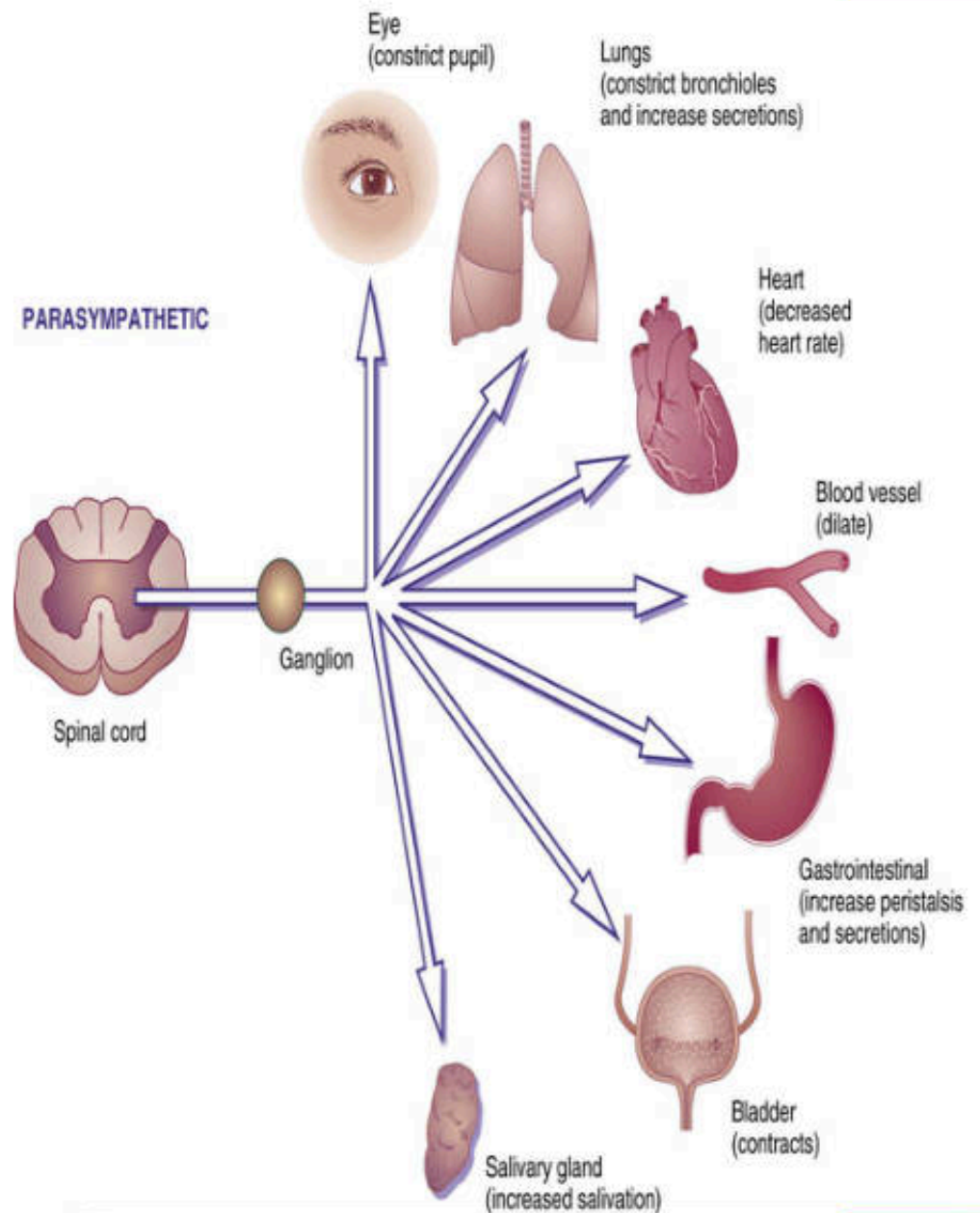
รศ.ดร. วิลาสินี หิรัญพานิช ซาโตะ

ภาควิชาเภสัชวิทยา คณะเภสัชศาสตร์
มหาวิทยาลัยมหิดล

Overview of Nervous System



Parasympathetic effects



Clinical manifestation of excessive cholinergic effects

DUMBBELLS



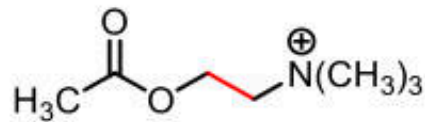
- D – Defecation (diarrhea)
- U - Urination (peeing)
- M - Miosis (constriction of the pupil of the eye), muscle spasm
- B - Bradycardia (slow heart beat)
- B - Bronchospasm (difficulty breathing)
- E – Emesis (excite skeletal muscle and CNS, vomiting)
- L - Lacrimation (tearing)
- L - Lethargy (fatigue)
- S - Salivation (excessive drooling), sweating

Cholinergic transmission

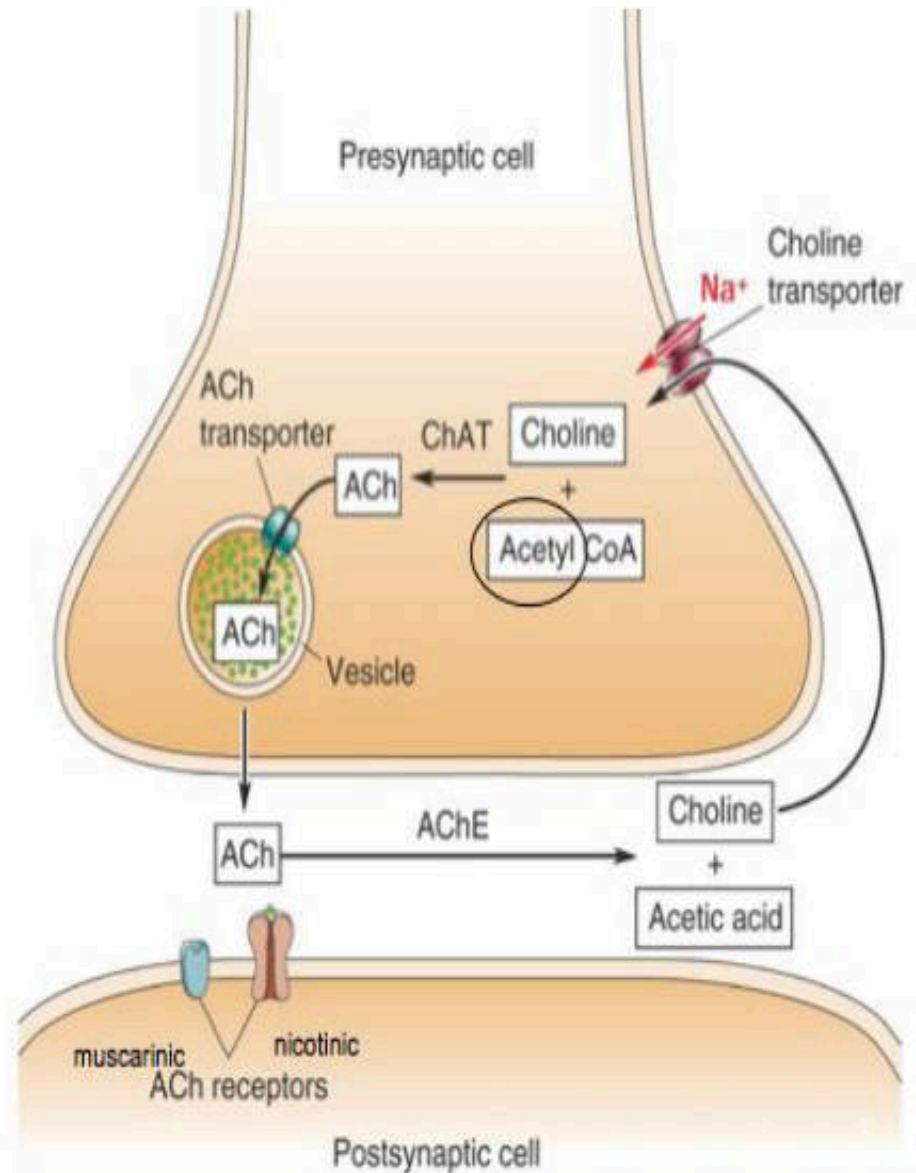
Neurotransmitter of
Parasympathetic
Nervous System



Acetylcholine
(ACh)



Cholinergic receptors: nicotinic (nAChR) and muscarinic (mAChR) subtypes



Degradation of ACh

- **1. Acetylcholinesterase (AChE)**
 - Most located in synapses and found in other tissues
- **2. Butyrylcholinesterase (BuChE) or pseudocholinesterase**
 - Low specificity on ACh
 - Found in blood, plasma, liver & many tissues
 - Not particularly associated with cholinergic synapses
 - Together with AChE, keeps acetylcholine concentration in plasma nearly undetectable

Muscarinic receptors

Muscarinic Receptors (mAChR) → G-protein-linked transmembrane receptors

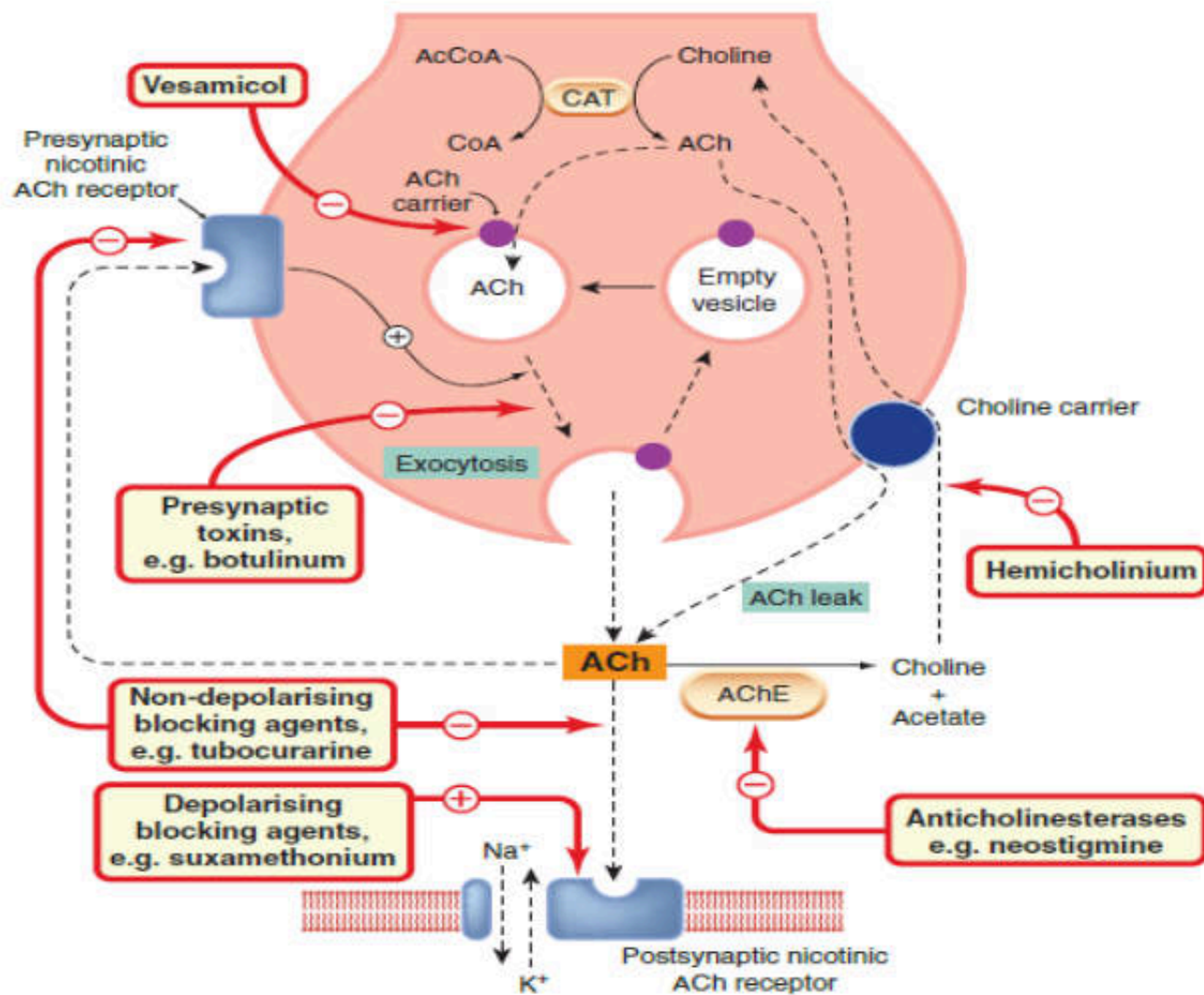
	M ₁ ('neural')	M ₂ ('cardiac')	M ₃ ('glandular/ smooth muscle')	M ₄	M ₅
Main locations	Autonomic ganglia (including intramural ganglia in stomach) Glands: salivary, lacrimal, etc. Cerebral cortex	Heart: atria CNS: widely distributed	Exocrine glands: gastric (acid-secreting parietal cells), salivary, etc. Smooth muscle: gastrointestinal tract, eye, airways, bladder Blood vessels: endothelium	CNS	CNS: very localised expression in substantia nigra Salivary glands Iris/ciliary muscle
Cellular response	↑ IP ₃ , DAG Depolarisation Excitation (slow epsp) ↓ K ⁺ conductance	↓ cAMP Inhibition ↓ Ca ²⁺ conductance ↑ K ⁺ conductance	↑ IP ₃ Stimulation ↑ [Ca ²⁺] _i	↓ cAMP Inhibition	↑ IP ₃ Excitation
Functional response	CNS excitation (? improved cognition) Gastric secretion	Cardiac inhibition Neural inhibition Central muscarinic effects (e.g. tremor, hypothermia)	Gastric, salivary secretion Gastrointestinal smooth muscle contraction Ocular accommodation Vasodilatation	Enhanced locomotion	Not known

Cholinergic receptors (cont.)

Nicotinic receptors (nAChR): Ligand-gated channels

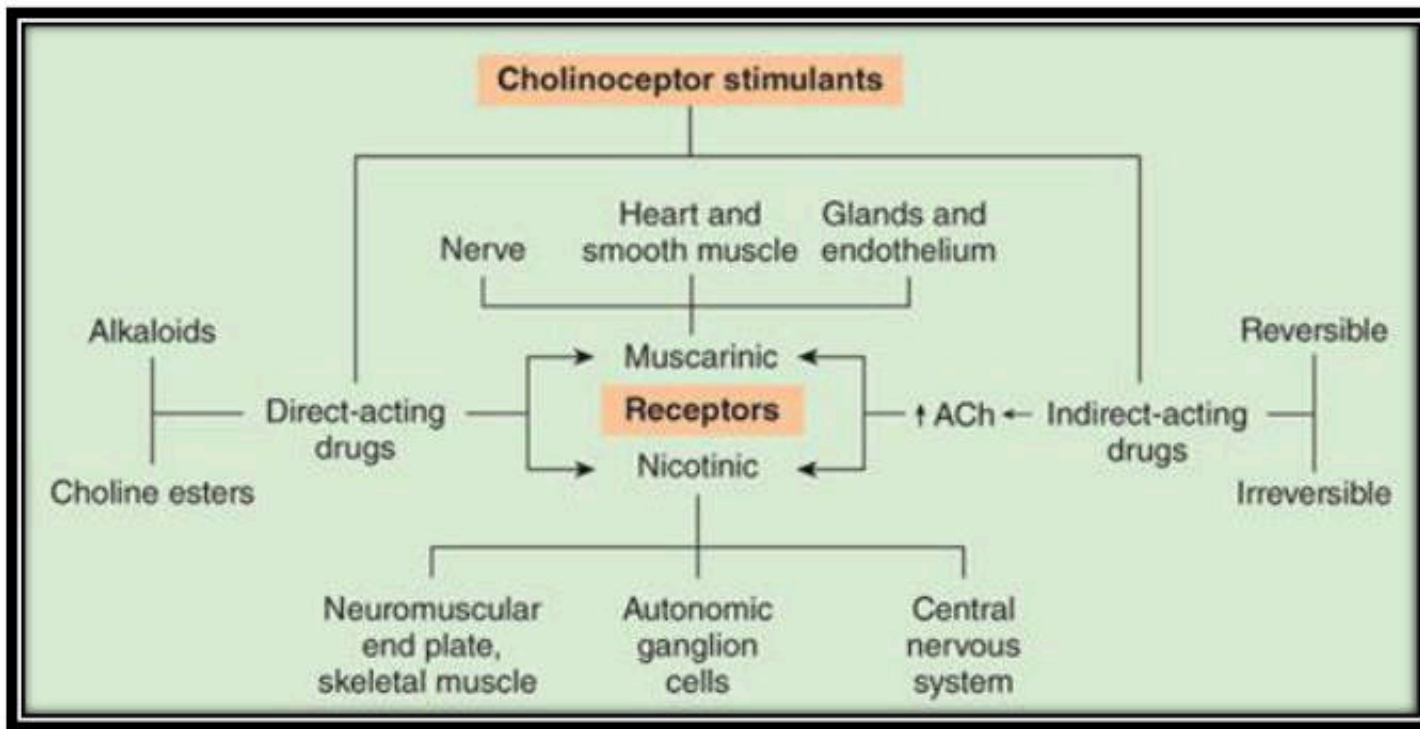
RECEPTOR (Primary Receptor Subtype) ^a	MAIN SYNAPTIC LOCATION	MEMBRANE RESPONSE	MOLECULAR MECHANISM	AGONISTS	ANTAGONISTS
Skeletal Muscle (N_m) (α_1) ₂ $\beta_1\epsilon\delta$ adult (α_1) ₂ $\beta_1\gamma\delta$ fetal	Skeletal neuromuscular junction (postjunctional)	Excitatory; end-plate depolarization; skeletal muscle contraction	Increased cation permeability (Na ⁺ ; K ⁺)	ACh Nicotine Succinylcholine	Atracurium Vecuronium <i>d</i> -Tubocurarine Pancuronium α -Conotoxin α -Bungarotoxin
Peripheral Neuronal (N_n) (α_3) ₂ (β_4) ₃	Autonomic ganglia; adrenal medulla	Excitatory; depolarization; firing of postganglion neuron; depolarization and secretion of catecholamines	Increased cation permeability (Na ⁺ ; K ⁺)	ACh Nicotine Epibatidine Dimethylphenyl- piperazinium	Trimethaphan Mecamylamine
Central Neuronal (CNS) (α_4) ₂ (β_4) ₃ (α -btox- insensitive)	CNS; pre- and postjunctional	Pre- and post- synaptic excitation Prejunctional control of transmitter release	Increased cation permeability (Na ⁺ ; K ⁺)	Cytosine, epibatidine Anatoxin A	Mecamylamine Dihydro- β - erythroline Erysodine Lophotoxin
(α_7) ₅ (α -btox- sensitive)	CNS; Pre- and post-synaptic	Pre- and post- synaptic excitation Prejunctional control of transmitter release	Increased permeability (Ca ²⁺)	Anatoxin A	Methyllycaconitine α -Bungarotoxin α -Conotoxin Iml

Events and sites of drug action at a nicotinic cholinergic synapse



Parasympathomimetic drugs (Cholinomimetic Drugs)

↑ activity of the parasympathetic nervous system

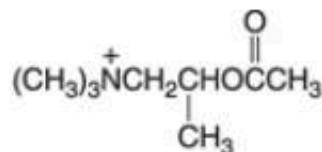


Direct acting parasympathomimetic drugs

1. Choline esters
 - ACh, Methacholine, Bethanicol, Carbachol
2. Natural or synthetic alkaloids
 - Naturals: Pilocarpine, Muscarine, nicotine (Nicotinia tabaum), lobeline Indian tobacco (Lobelia sp.), Arecoline
 - Synthetic alkaloids: oxotremorine, cevimeline



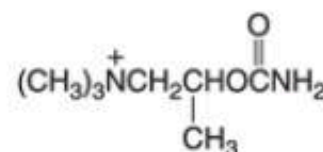
ACETYLCHOLINE



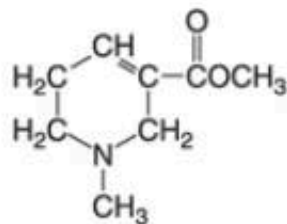
METHACHOLINE



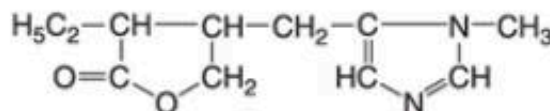
CARBACHOL



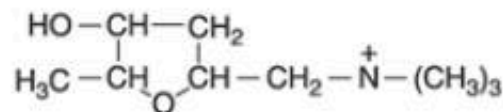
BETHANECHOL



ARECOLINE



PILOCARPINE



MUSCARINE

Pharmacological properties of Choline ester and natural alkaloids

Muscarinic agonists	ความไวในการถูกทำลายด้วยเอนไซม์ Ach esterase	Muscarinic activity				การถูกต้านฤทธิ์โดยยา atropine	Nicotinic activity	ประโยชน์ทางคลินิก
		หัวใจ	GI tract	กระเพาะปัสสาวะ	ตา			
Acetylcholine	+++	++	++	++	+	+++	++	Acetylcholine: used only topically for the induction of miosis during ophthalmologic surgery (1% solution)
Methacholine	+	+++	++	++	+	+++	+	-
Carbachol	-	+	+++	+++	++	+	+++	ใช้หยอดม่านตา
Bethanechol	-	+/-	+++	+++	++	+++	-	กระตุ้นการหดตัวลำไส้
Muscarine	-	++	+++	+++	++	+++	-	-
Pilocarpine	-	+	+++	+++	++	+++	-	ลดความดันในดวงตา

Pharmacological effects “Muscarinic Effects”

Systems	Effects
Cardiovascular System	Vasodilation, ↓ heart rate (negative chronotropic effect), ↓ conduction velocity in the atrioventricular (AV) node (negative dromotropic effect), ↓ force of cardiac contraction (negative inotropic effect)
Respiratory Tract	bronchoconstriction, increase tracheobronchial secretion (M3)
Urinary Tract	detrusor muscle contraction, increase voiding pressure, and ureteral peristalsis (M2)
GI Tract	increases tone, amplitude of contractions, and secretory activity of the stomach and intestine (M2,M3)
CNS Effects	cognitive function, motor control, appetite regulation, nociception
Eyes	Miosis (accommodation for near vision by contracting the ciliary muscle)
Secretion and Sweat gland	↑ sweat, ↑secretion, ↑saliva secretion

Pharmacological effects (cont.)

“Nicotinic Effects”

- **Autonomic ganglia**
 - Stimulate sympathetic neuron
 - Stimulate parasympathetic neuron
- **Neuromuscular junction (NMJ):**
 - Muscle contraction
 - Muscle fasciculation
 - Paralysis (depolarized blockade)
- CNS:
 - Stimulation release of neurotransmitters (glutamate, serotonin, GABA, dopamine, NE)

Contraindications, Precautions, and Adverse Effects

Contraindications

- Asthma
- Chronic obstructive pulmonary disease
- Urinary or GI tract obstruction
- Acid-peptic disease
- Cardiovascular disease accompanied by bradycardia, hypotension
- Hyperthyroidism (muscarinic agonists may precipitate atrial fibrillation in hyperthyroid patients)

Adverse effects

- Diaphoresis (ภาวะหลังเหงื่อมาก)
- Diarrhea
- Abdominal cramps
- Nausea/vomiting
- GI side effects
- A sensation of tightness in the urinary bladder
- Difficulty in visual accommodation
- Hypotension

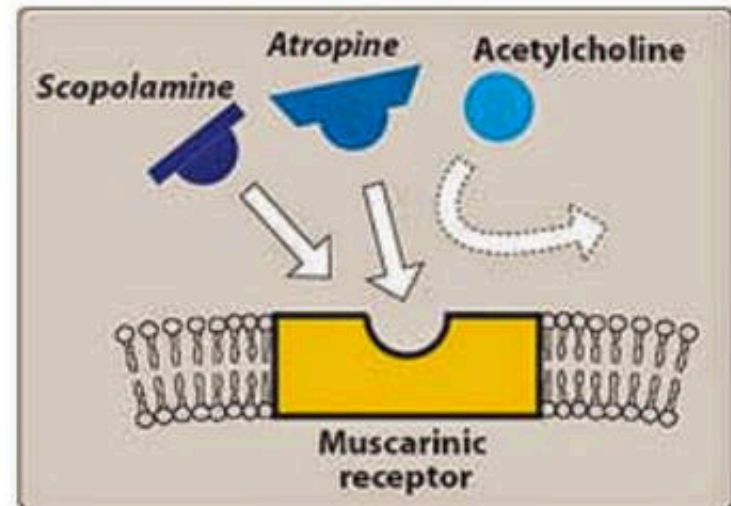
Toxicology and treatments

Symptoms

- Exaggeration of parasympathomimetic effects

Treatments

- Use parenteral administration of atropine (antimuscarinic agent)
- Support the respiratory and cardiovascular systems



Clinical uses of muscarinic agonists and related drugs

- **Pilocarpine:**
 - Topically in ophthalmology for the treatment of glaucoma and the induction of miosis (0.5%–6% solution and ocular insert)
 - Treatment of xerostomia: after head and neck radiation and Sjögren syndrome
- **Carbachol:**
 - topically in ophthalmology for the treatment of glaucoma and the induction of miosis
 - Miotics agent, cataract surgery
- **Cevimeline:**
 - Selective muscarinic agonist at **M1 and M3 receptors** on lacrimal and salivary gland epithelia
 - Long-lasting sialogogic action and may have **fewer side effects and better patient compliance** than pilocarpine
 - increase salivation and lacrimal secretion in patients with dry mouth or eyes (Sjögren's syndrome)

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11.4 Drugs for treatment of glaucoma

1.	Glycerol	oral sol (hosp)	ก
2.	Acetazolamide	tab	ก
3.	Pilocarpine	eye drop (as hydrochloride or nitrate)	ก

Carbachol

sterile sol for intraocular use

ค 11.7. Ocular diagnostic
and peri-operative
preparations and
photodynamic treatment

เงื่อนไข : ใช้หัตถ์มาบดระหว่างการทำผ่าตัด

2. Indirect acting drugs [AChE Inhibitors]

(Anticholinesterase Agents
or Cholinesterase inhibitor)

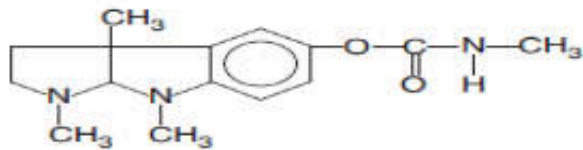


**1. Reversible
cholinesterase
inhibitors**

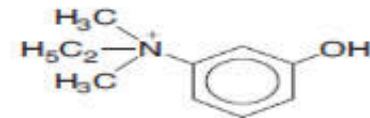


**2. Irreversible
cholinesterase
inhibitors**

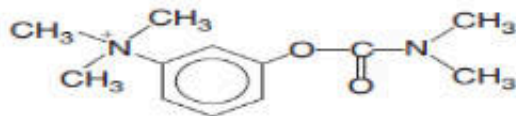
Reversible cholinesterase inhibitors



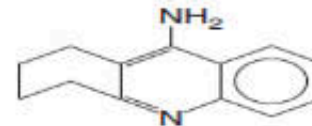
PHYSOSTIGMINE



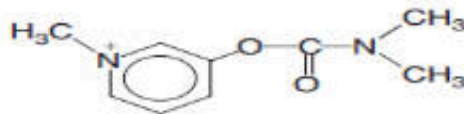
EDROPHONIUM



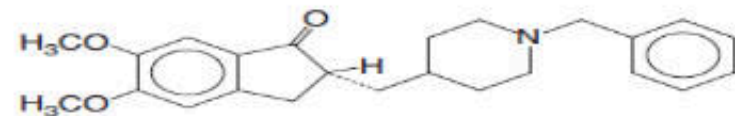
NEOSTIGMINE



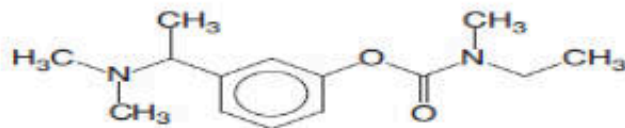
TACRINE



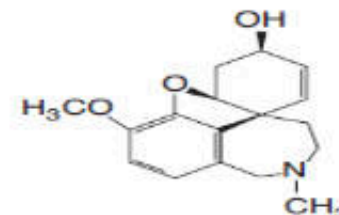
PYRIDOSTIGMINE



DONEPEZIL



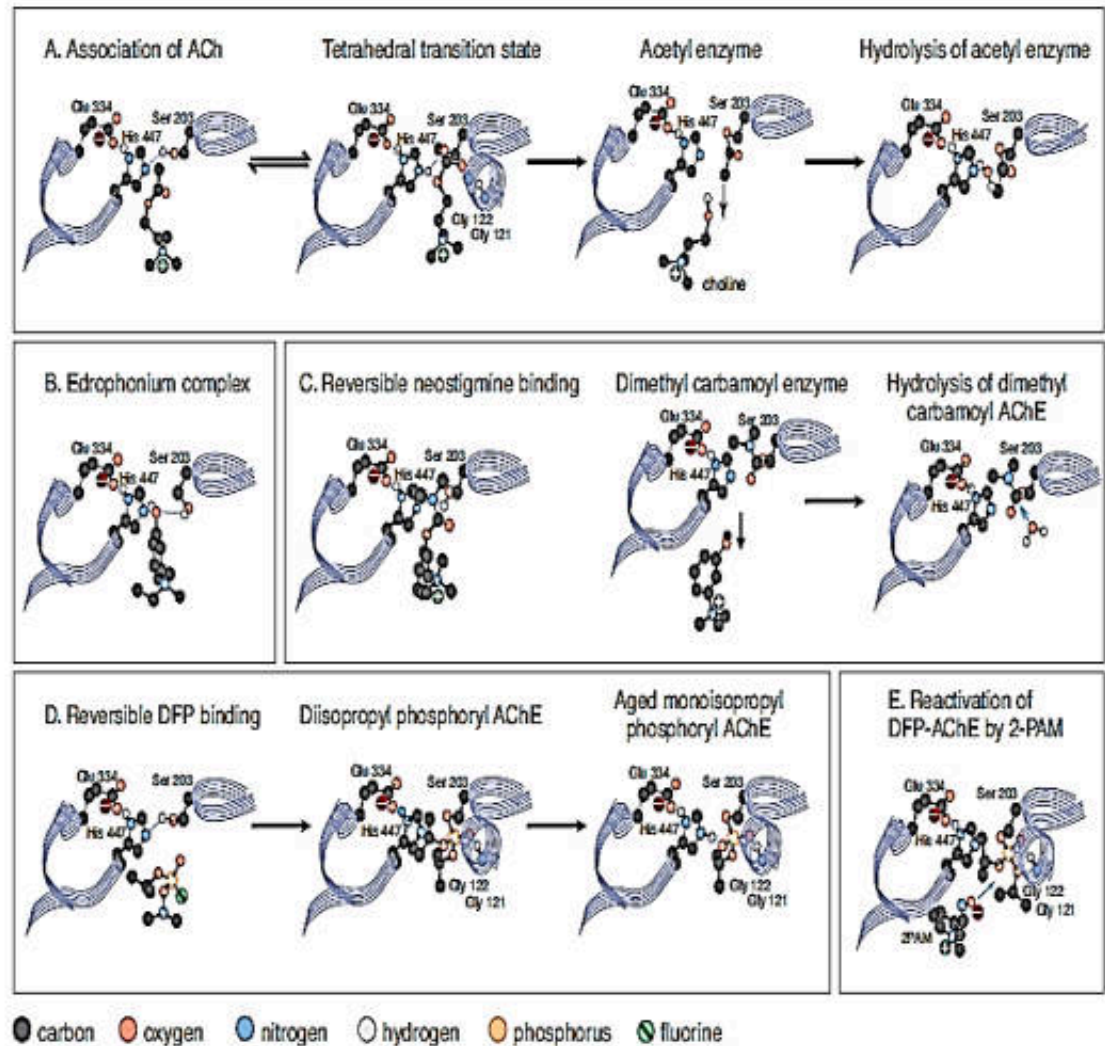
RIVASTIGMINE



GALANTAMINE

1. Short-acting cholinesterase inhibitors

- **Edrophonium**
- Structure: quaternary amine
- Binds reversibly to the AChE active center
- Brief duration of action because its quaternary structure facilitates renal elimination
- Short acting (5-15 min)
- Use: “Diagnosis of myasthenia gravis”



2. Medium-duration cholinesterase inhibitors

carbamyl esters interact with the anionic site of AChE

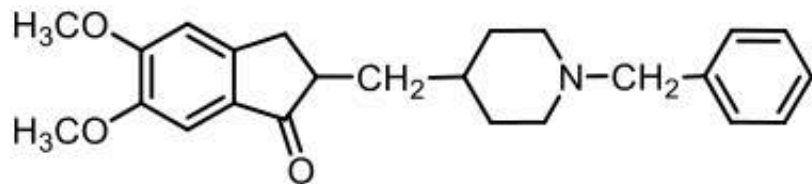


“carbamylated enzyme” (much slower to be hydrolyzed)

- **Duration of action: 30 min - 6 hr.**
- **Example of drugs:**
 - **Neostigmine:** target mainly NMJ, treat myasthenia gravis (oral form)
 - **Pyridostigmine:** target mainly NMJ (longer duration than Neostigmine), treat myasthenia gravis (oral forms), reverse neuromuscular block (*i.v.*)
 - **Physostigmine:** post-ganglionic parasympathetic junctions (use for glaucoma; eye drop)
- **Other reversible AChE Inhibitors:** **Alzheimer's Disease: Donepezil, Rivastigmine, Galantamine**

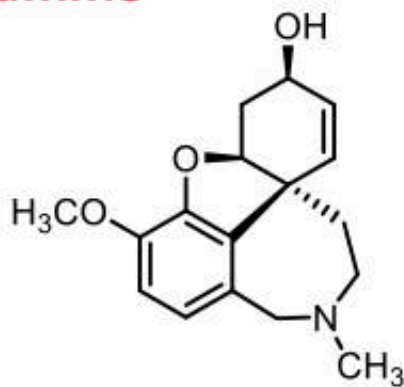
Other reversible AChEI: Alzheimer's Disease

➤ Donepezil



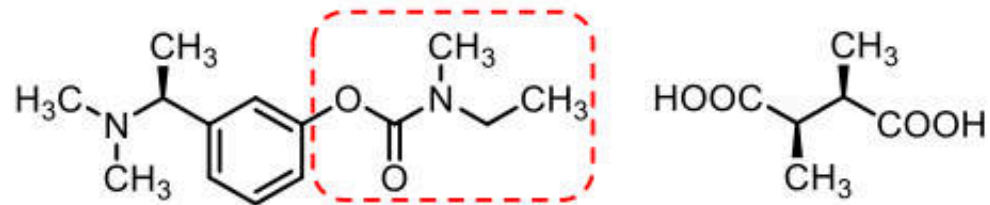
- Non-classical cholinesterase inhibitor
- **Selective to AChE** > BuChE

➤ Galantamine



- Not inhibit BuChE
- Allosterically bind to nAChR (dual cholinergic action)

➤ Rivastigmine tartrate



- Slow hydrolysis of carbamylated enzyme
- Low potential of hepatotoxicity & DRUG Interaction

Toxicity

Muscarinic receptor stimulation:

- nausea, vomiting, diarrhea, salivation, sweating, vasodilation, bradycardia, bronchial constriction

Nicotinic receptor stimulation:

- Twitching and generalized spasm
- Depolarization blockade lead to respiratory paralysis

Clinical uses

- **Eyes:** glaucoma, miosis, accommodation (Physostigmine)
- **Myasthenia gravis:** Edrophonium, Neostigmine, Pyridostigmine
- **Mild to moderate Alzheimer's disease:** donepezil, rivastigmine, galantamine
- **Antimuscarinic drug intoxication (Neostigmine)**

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10.2. Drugs used in neuromuscular disorders

10.2.1. Drugs which enhance neuromuscular transmission

1	Pyridostigmine bromide		tab	ก
2	Neostigmine methylsulfate		sterile sol	ข

15.1.5. Anticholinesterases and antimuscarinic drugs used in anesthesia

1	Atropine sulfate		sterile sol	ก
2	Neostigmine methylsulfate		sterile sol	ก
3	Edrophonium chloride ยากำพรั้า		sterile sol (เฉพาะ 10 mg/mL)	ง
4	Glycopyrronium bromide		sterile sol	ง
5	Sugammadex		sterile sol (เฉพาะ 100 mg/mL) (2 mL)	ง

Glycopyrronium bromide

sterile sol

ง

15.1.5

Anticholinesterases and antimuscarinic drugs used in anaesthesia

เงื่อนไข :

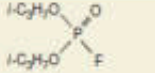
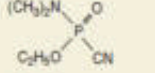
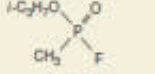
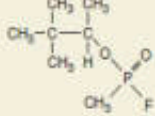
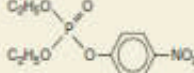
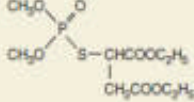
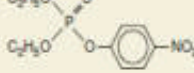
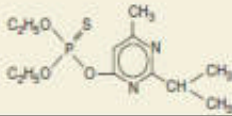
1. ใช้เป็นยาทางเลือกแทนยา atropine โดยใช้ร่วมกับยา neostigmine ในการแก้ฤทธิ์ยาหย่อนกล้ามเนื้อในผู้ป่วยสูงอายุ หรือผู้ป่วยที่มีปัญหาหัวใจเต้นเร็ว
2. ใช้ลดเสมหะหรือน้ำลายในการดมยาสลบกรณีที่จะทำหัตถการเกี่ยวกับทางเดินหายใจ
3. ใช้ลดเสมหะในผู้ป่วยระยะสุดท้าย (end-of-life)

3. Irreversible cholinesterase inhibitors (Organophosphate agents)

- Irreversible anticholinesterase by phosphorylation
- Phosphorelated enzyme is very stable!
- Wait for new enzyme is synthesized
- Depress enzymatic activity persist for weeks or months
- The phosphorylated enzyme may undergo a process called “Aging”

Examples:

- Dyflos (isofluorophate)
- Ecothiopate (insecticide)
- Parathion (insecticide)
- Malathion (insecticide)
- Sarin (gas)

GROUP	STRUCTURAL FORMULA	COMMON, CHEMICAL, AND OTHER NAMES	COMMENTS
A		DFP; Isofluorophate; diisopropyl fluorophosphate	Potent, irreversible inactivator
		Tabun Ethyl N-dimethylphosphoramidocyanidate	Extremely toxic “nerve gas”
		Sarin (GB) Isopropyl methylphosphonofluoridate	Extremely toxic “nerve gas”
		Soman (GD) Pinacolyl methylphosphonofluoridate	Extremely toxic “nerve gas”; greatest potential for irreversible action/ rapid aging
B		Paraoxon (MINTACOL), E 600 O,O-Diethyl O-(4-nitrophenyl)-phosphate	Active metabolite of parathion
		Malaoxon O,O-Dimethyl S-(1,2-dicarboxyethyl)-phosphorothioate	Active metabolite of malathion
C		Parathion O,O-Diethyl O-(4-nitrophenyl)-phosphorothioate	Agricultural insecticide, resulting in numerous cases of accidental poisoning; phased out in 2003.
		Diazinon, Dimpylate O,O-Diethyl O-(2-isopropyl-6-methyl-4-pyrimidinyl) phosphorothioate	Insecticide; use limited to non-residential agricultural settings

Toxic reactions irreversible cholinesterase inhibitors

- **Muscarinic effects:**

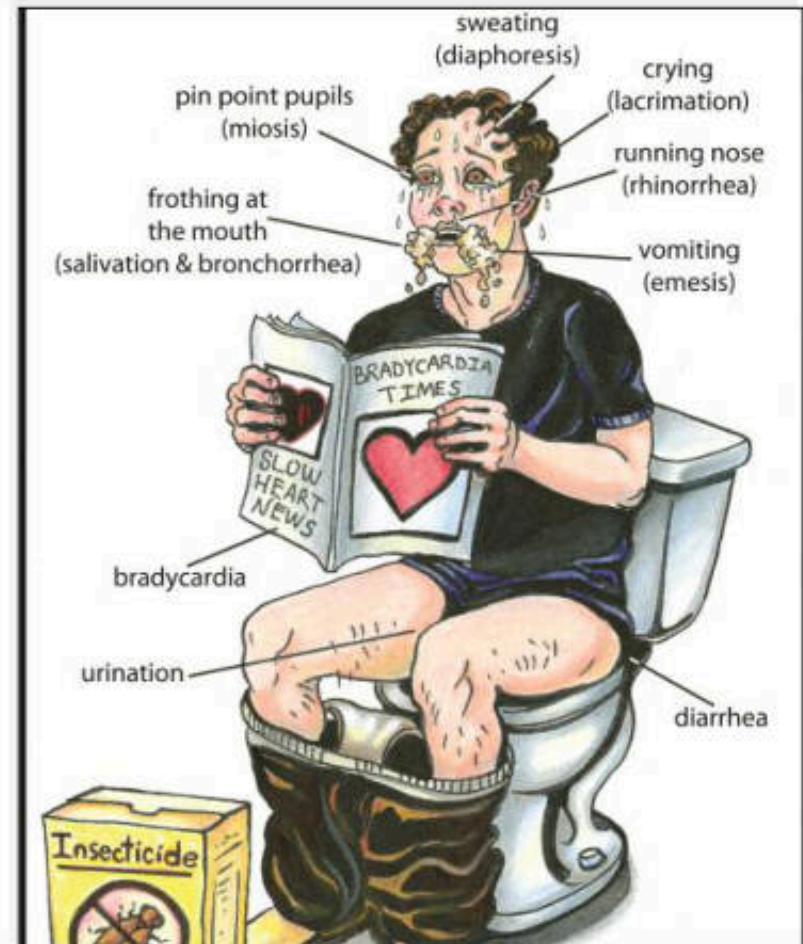
- Ocular: miosis, blurred vision, headache
- Secretions: lacrimal, salivary, bronchial...
- Smooth muscle contraction: bronchial, visceral
- Cardiovascular: bradycardia, hypotension

- **Nicotinic effects:**

- Skeletal muscle: fasciculation, weakness
- Cardiovascular: tachycardia, blood pressure instability

- **CNS effects:**

- Mixed excitatory and depressant effects
- Respiratory and circulatory collapse

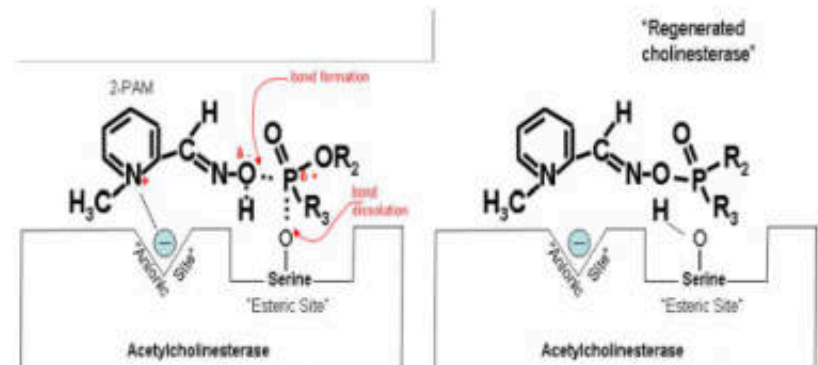


Treatment of Toxicity

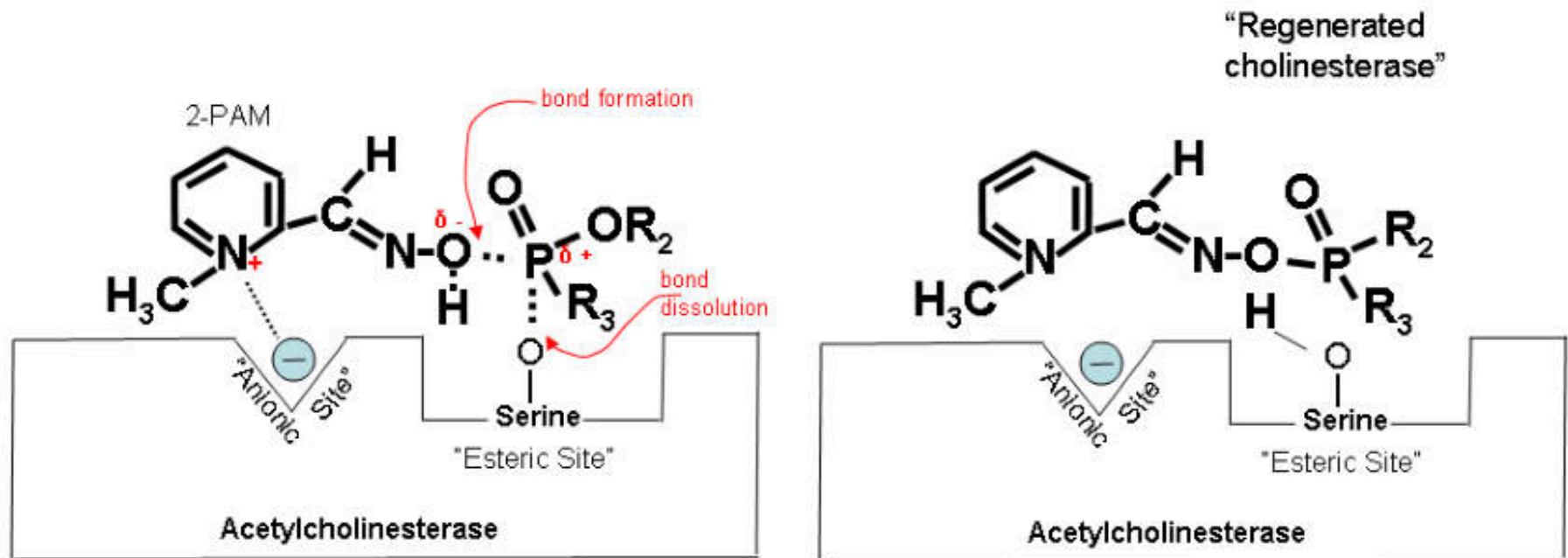
- Decontamination:
activated charcoal
- Supportive care:
airway, respiratory
support
- Antidotes
 - **Atropine**: anticholinergic
drug
 - **Pralidoxime (2-PAM)**
 - **Obidoxime**
 - **Diacetylmonoxime (DAM)**

Cholinesterase Reactivators (Antidotes)

- Pralidoxime (2-PAM) prevents aging and regenerates AChE
Regeneration of free enzyme
(Within a short period of time after
enzyme phosphorylation)



Pralidoxime (2-PAM) prevents aging and regenerates AChE



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16.3 Pesticide & herbicide poisoning

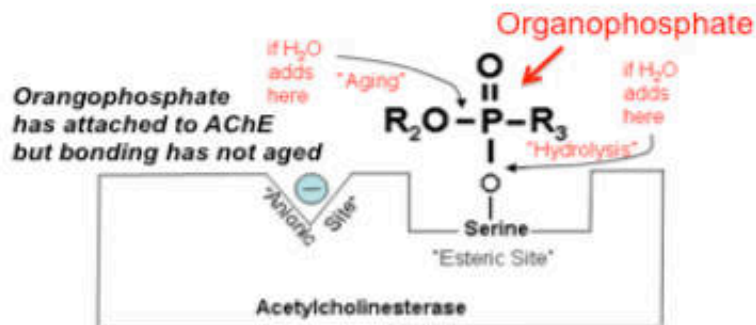
1	Atropine sulfate		sterile sol	ก
2	Pralidoxime chloride	2-PAM	sterile pwdr	ก
3	Cyclophosphamide		sterile pwdr	ค

เงื่อนไข : 2-PAM ใช้บำบัดพิษที่เกิดจาก organophosphates

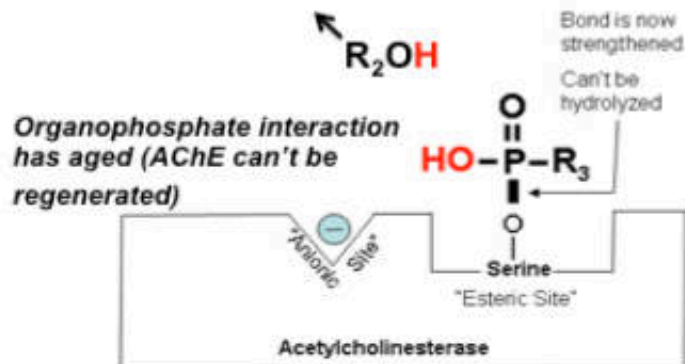
Aging

- breaking of one of the oxygen-phosphorus bonds of inhibitor and further strengthens the phosphorus-enzyme bond

Organophosphate Aging – chemical stabilization of phosphate bond to AChE occurs over time



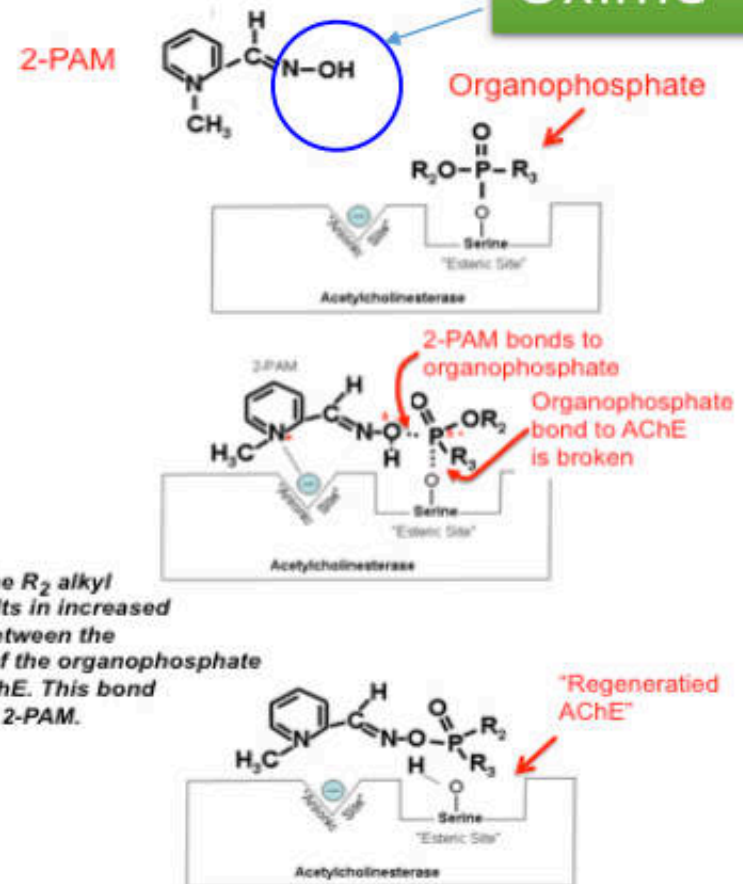
The rate of aging is unique for each organophosphate compound, and can occur over minutes to days depending on the agent



The departure of the R_2 alkyl group (aging) results in increased electron sharing between the phosphate group of the organophosphate & the serine on AChE. This bond can't be broken by 2-PAM.

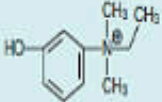
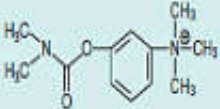
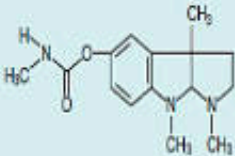
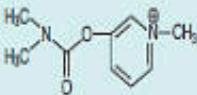
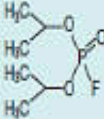
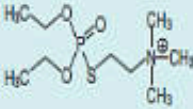
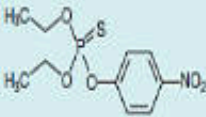
Pralidoxime (2-PAM) prevents aging & regenerates AChE

Oxime



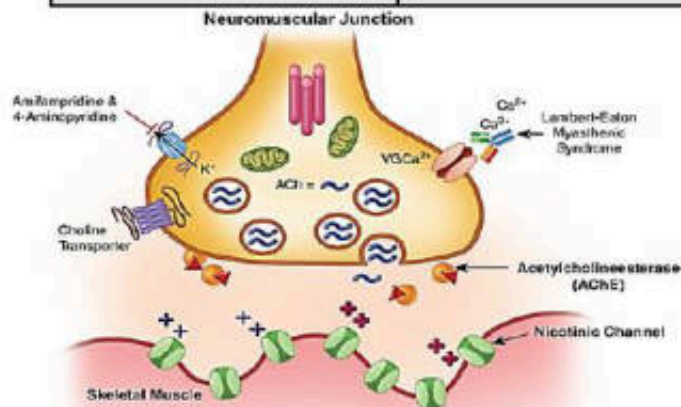
must be given early before "aging process"

Summary of AChE Inhibitors drugs

Drug	Structure	Duration of action	Main site of action	Notes
Edrophonium		Short	NMJ	Used mainly in diagnosis of myasthenia gravis Too short-acting for therapeutic use
Neostigmine		Medium	NMJ	Used intravenously to reverse competitive neuromuscular block Used orally in treatment of myasthenia gravis Visceral side effects
Physostigmine		Medium	P	Used as eye drops in treatment of glaucoma
Pyridostigmine		Medium	NMJ	Used orally in treatment of myasthenia gravis Better absorbed than neostigmine and has longer duration of action
Dyflon		Long	P	Highly toxic organophosphate, with very prolonged action Has been used as eye drops for glaucoma
Ecothiopate		Long	P	Used as eye drops in treatment of glaucoma Prolonged action; may cause systemic effects
Parathion		Long	-	Converted to active metabolite by replacement of sulfur by oxygen Used as insecticide but also causes poisoning in humans

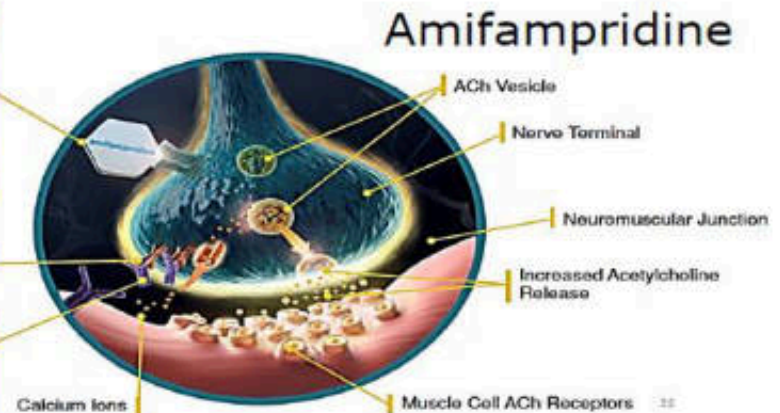
New drugs for antidote “Acetylcholine Release Modulators”

Drug	Mechanism of action	Effects	Clinical applications	Pharmacokinetics/Toxicities/Interactions
Amifampridine and 4-aminopyridine	- Potassium channel antagonist - inhibiting the K^+ efflux and slowing repolarization	- Increases the excitability of the presynaptic neuron in the neuromuscular junction - resulting in increased ACh release	Lambert-Eaton myasthenic syndrome -antibodies are produced that interfere with proper voltage-gated Ca^{2+} ($VGCa^{2+}$) channel function on the presynaptic nerve of the neuromuscular junction and reduce ACh release	- Oral, duration ~2.5 h - Toxicity: seizure activity - may prolong cardiac QT interval - paresthesias - numbness - insomnia



Amifampridine blocks potassium channels, allowing calcium channels to stay open longer. Open channels let more Ca^{2+} inside the nerve terminal, increasing acetylcholine (ACh) release. Increased ACh binding to muscle ACh receptors restores lost muscle strength

Autoantibodies block Ca^{2+} channels, preventing Ca^{2+} from entering nerve terminals and reducing ACh release. This results in reduced ACh in NMJ and muscle weakness



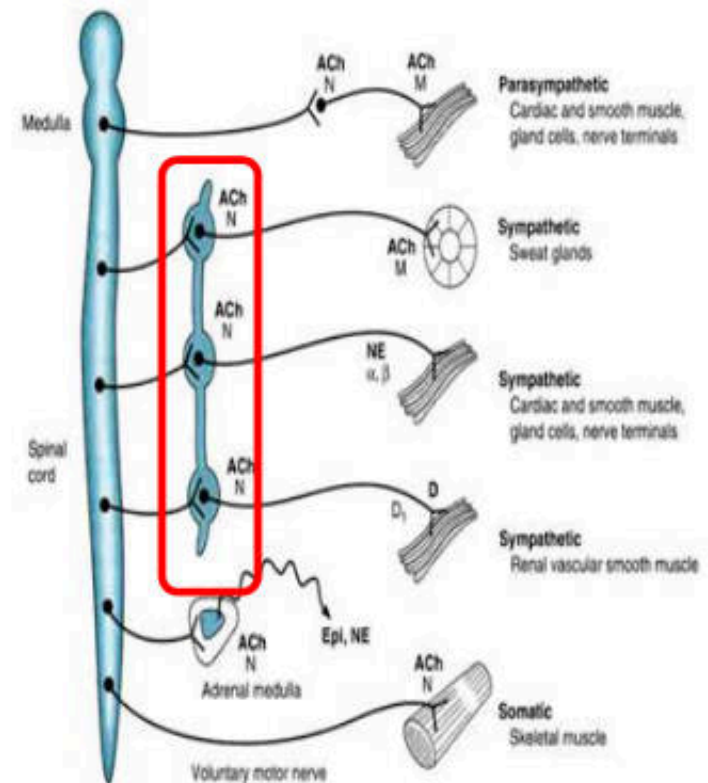
Ref: Facebook Infectious ง่ายนิดเดียว Sept 28, 2021

- มีเด็กอายุ 1 เดือน 1 วัน น้ำหนัก 3 กก. เอง มา รพ. ด้วยอาการหนัก เลยมารตรวจห้องฉุกเฉิน มีอาการ หอบเหนื่อย ตัวโยน ชีต ตัวเขียว น้ำลายมาก เสมหะมาก **hypersecretion** ม่านตาตึบแคบ ชีพจรเต้นช้า ภาวะวิกฤต นู๋ที่ฉุกเฉิน ต้องรีบให้ออกซิเจน โชคดี พยาบาลทีมเวร วันนั้น แจ้งว่ามีเคสลักษณะแบบนี้ มาติดต่อกันในบ้านนี้ ก่อนหน้านั้นผู้ใหญ่ 3 ราย เคสเด็กเป็นเคสที่ 4 ที่มีอาการแบบเดียวกัน น้องหมอเวรที่ รพ.ชุมชนสงสัย ได้รับสารพิษ!!
- ซึ่งอาการดังกล่าวข้างต้น เข้าได้กับ สารกลุ่ม **organophosphate** คือยาฆ่าแมลง ได้ย้อนחקประวัติกลับไป ในบ้านมียาฆ่าแมลงและวางใกล้แหล่งน้ำ ซึ่งอาจปนเปื้อนในน้ำกินน้ำใช้ ทำให้คนในบ้านได้รับสารพิษ
- แพทย์ รีบให้การรักษาด้วยยาต้านฤทธิ์ ยากลุ่ม **organophosphate** ด้วยยา **Atropine 0.05 mg/kg** ทางเส้นเลือด i.v. (ส่วนยาอื่นๆ 2-PAM) กับช่วยทางเดินหายใจ ใส่ท่อช่วยหายใจ ให้ออกซิเจน หลังจากนั้นอาการเด็ก ดีขึ้น ชีตเจ็น หายใจหอบเหนื่อยน้อยลง เสมหะน้อยลง ตัวหายชิต และเขียว กลับมาแดงแล้ว ชีพจรเต้นเร็วขึ้น รอดแล้ว
- วินิจฉัย **Organophosphate toxicity**
- เตือนภัย สำหรับประชาชน ควรเก็บสารเคมีให้ไกลจากแหล่งน้ำกินน้ำใช้ ทำความสะอาดร่างกายหลังใช้สารเคมี



Ganglion-stimulating drugs

- Example: Nicotine
- Stimulate sympathetic and parasympathetic ganglia
 - tachycardia and increase of blood pressure
 - GI motility and secretions
 - increased bronchial, salivary and sweat secretions
- May be followed by depolarization block
- No therapeutic uses, except for nicotine (smoking cessation)



Nicotine

- Stimulation of nicotinic receptors
- **Mechanism of action:** stimulate N_n receptor at ganglion and adrenal medulla
- CNS stimulant: Increase alertness, facility of memory, cognitive function, reduce aggressive, suppress appetite
- Pleasure system → Addiction!
- Use to aid smoking cessation
- Clinical uses:
 - Nicotine replacement therapy (Gum, nicotine transdermal patch, nasal spray, inhaler, lozenges, sublingual tablets)

DIRECT-ACTING NICOTINIC AGONISTS				
Drug	Mechanism of action	Effects	Clinical applications	Pharmacokinetics/Toxicities/Interactions
*Nicotine	Agonist at both N_n and N_m receptors	• Activates autonomic postganglionic neurons (both sympathetic and parasympathetic) and skeletal muscle neuromuscular end plates • enters CNS and activates N_m receptors	• Medical use in smoking cessation • nonmedical use in smoking and in insecticides	• Oral gum, patch for smoking cessation • Toxicity: Acutely increased gastrointestinal (GI) activity, nausea, vomiting, diarrhea • increased blood pressure • high doses cause seizures • long-term GI and cardiovascular risk factor • Interactions: Additive with CNS stimulants
• Varenicline: Selective partial agonist at $\alpha 4\beta 2$ nicotinic receptors; used exclusively for smoking cessation				

แบบฝึกหัด

ทบทวน

- 1. Which enzyme is inhibited by indirect-acting parasympathomimetic drugs?
 - A) Monoamine oxidase
 - B) Acetylcholinesterase
 - C) Cyclooxygenase
 - D) Phosphodiesterase
- 2. Which of the following drugs is used as an antidote for organophosphate poisoning?
 - A) Pralidoxime
 - B) Bethanechol
 - C) Pilocarpine
 - D) Physostigmine
- 3. Which parasympathomimetic drug is commonly used to stimulate bladder contraction in cases of urinary retention?
 - A) Atropine
 - B) Bethanechol
 - C) Scopolamine
 - D) Ipratropium
- 4. Which of the following is a contraindication for the use of parasympathomimetic drugs?
 - A) Glaucoma
 - B) Urinary retention
 - C) Bradycardia
 - D) Hypertension

Example of case study

- Miss A underwent a total knee replacement surgery three days ago. She now presents with an inability to urinate for the past 8 hours despite feeling a strong urge. She describes discomfort and bloating in her lower abdomen.
- **Investigations:**
- **Bladder Ultrasound:** Bladder volume of 700 mL, indicating urinary retention
- **Urinalysis:** No signs of infection, normal results

1. Which drug is appropriate for the treatment?

- A. Pilocarpine
- B. Bethanechol
- C. Malathion
- D. Rivastigmine

2. Following the answer in (1) what is the mechanism of action?

- A. Direct acting sympathomimetic drugs
- B. Indirect acting sympathomimetic drugs
- C. Direct acting parasympathomimetic drugs
- D. Indirect acting parasympathomimetic drugs

3. What is the possible side effects

- A. Hypertension
- B. Nasal congestion
- C. Diarrhea

2. Parasympatholytics Drugs (Anticholinergics Drugs)

- Cholinergic blockers/antagonists or Acetylcholine blocking agents
 1. Ganglionic blocking drugs
 2. Antimuscarinic drugs (Anticholinergic drugs)
 3. Neuromuscular blocking drugs (NMB)

1. Ganglion blocking agents

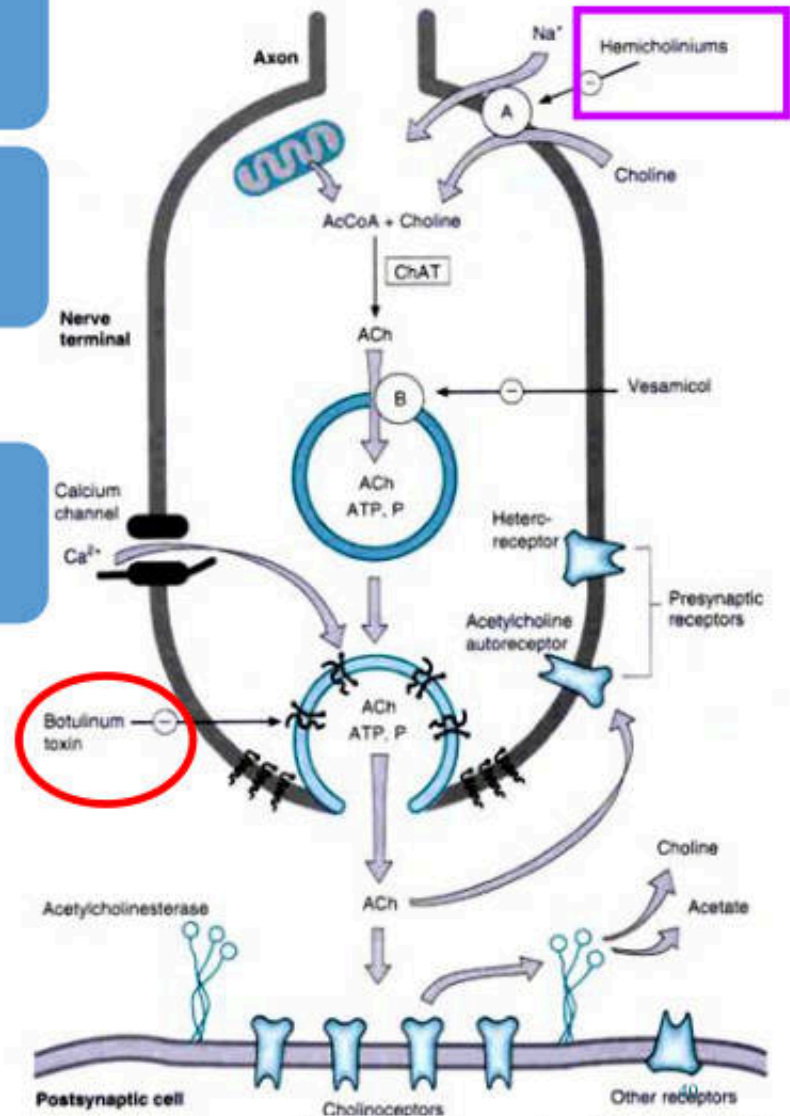
Block all autonomic ganglia and enteric ganglia

Inhibit N_n receptor (nAChR ganglionic type)

- Examples: Hexamethonium, trimethaphan, pentonium, Mecamylamine

Clinical uses:

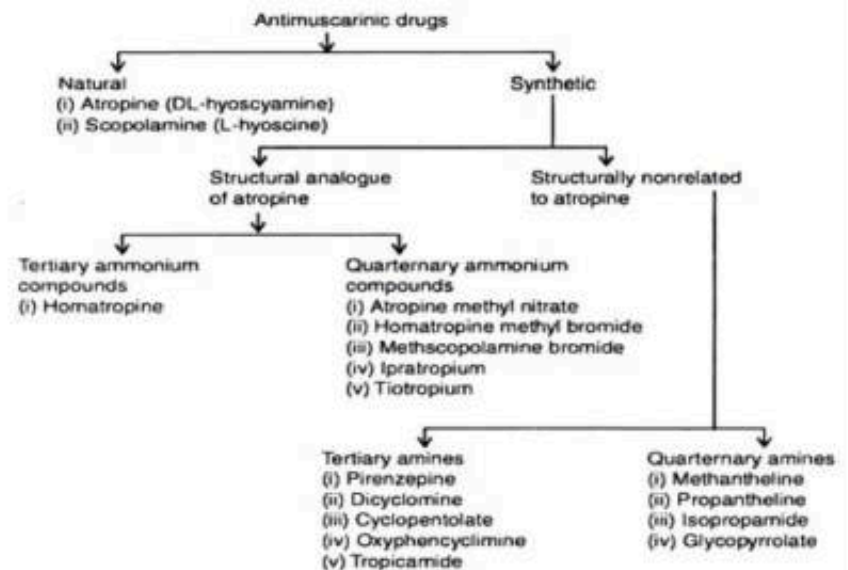
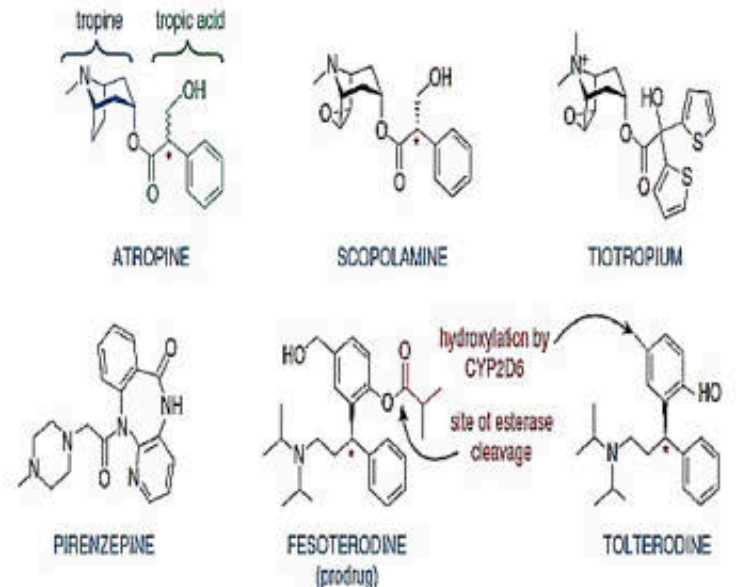
- **Botulinum toxin:** blocks nerve-stimulated release of ACh
- U.S. FDA approved for strabismus: crossed eyes)
อาการตาเหล่ (strabismus: crossed eyes) และ blepharospasm (ตาปิดเกร็ง, การกระพริบตาที่ไม่สามารถควบคุมได้)
- **Hemicholinium:** blocks transport of choline into neuron
- **Mecamylamine hydrochloride**



2. Antimuscarinic Drugs (Muscarinic antagonists)

Competitive inhibition of
muscarinic receptor

- 1. Natural alkaloids
 - Atropine extracted from belladonna alkaloids
 - Prototype of antimuscarinic drugs
- 2. Synthetic atropine like drugs
 - Tertiary amines
 - Quaternary amines



Muscarinic antagonists

Compound	Pharmacological properties	Notes
Atropine	Non-selective antagonist Well absorbed orally CNS stimulant	Belladonna alkaloid Main side effects: urinary retention, dry mouth, blurred vision Dicycloverine (dicyclomine) is similar and used mainly as an antispasmodic agent
Hyoscine	Similar to atropine CNS depressant	Belladonna alkaloid (also known as scopolamine) Causes sedation; other side effects as atropine
Hyoscine butylbromide	Similar to atropine but poorly absorbed and lacks CNS effects Significant ganglion-blocking activity	Quaternary ammonium derivative Similar drugs include atropine methonitrate, propantheline
Tiotropium	Similar to atropine methonitrate Does not inhibit mucociliary clearance from bronchi	Quaternary ammonium compound Ipratropium similar
Tropicamide	Similar to atropine May raise intraocular pressure	–
Cyclopentolate	Similar to tropicamide	–
Darifenacin	Selective for M ₃ receptors	Urinary incontinence Few side effects

Pharmacological actions

ORGANS	EFFECTS
Eye	dilate the pupil (mydriasis) and paralyze accommodation (cycloplegia), photophobia, lost of accommodation (near objects appear blurred, dry eye)
GI tracts	↓gastric, motilitygastric secretion, ↓ tone and amplitude of peristaltic contraction
Respiratory tracts	secretions of nose, pharynx, and bronchi, Dry the mucous membrane, Bronchodilation, Relaxation of smooth muscles
CVS	Heart: Increase heart rate (inh. M2 receptor of SA node), Increase AV conduction velocity Blood pressure: Not much affect on Blood pressure Blood vessel: “Atropine flush” vasodilation (high dose)
CNS	minimal effects (normal dose), central excitation followed by respiratory Failure (toxic doses)
Secretions	decrease secretion, dry mouth
Sweat glands and temperature	inhibit the activity of sweat glands, ↑ body temperature
Smooth Muscle	Urinary Tract (↓normal tone and amplitude of contractions of the ureter and bladder) Biliary Tract (antispasmodic action on the gallbladder and bile ducts)

Adverse Drug reactions and Contraindications

Xerostomia (ภาวะปากแห้ง)

Constipation

Blurred Vision

Dyspepsia

Cognitive Impairment

Contraindications:

- Urinary tract obstruction, GI obstruction
- Uncontrolled angle-closure glaucoma
- Patients with benign prostatic hyperplasia

Effects of Atropine in Relation to Dose

DOSE (mg)	EFFECTS
0.5	Slight cardiac slowing; some dryness of mouth; inhibition of sweating
1	Definite dryness of mouth; thirst; acceleration of heart, sometimes preceded by slowing; mild dilation of pupils
2	Rapid heart rate; palpitation; marked dryness of mouth; dilated pupils; some blurring of near vision
5	Above symptoms marked; difficulty in speaking and swallowing; restlessness and fatigue; headache; dry, hot skin; difficulty in micturition; reduced intestinal peristalsis
≥10	Above symptoms more marked; pulse rapid and weak; iris practically obliterated; vision very blurred; skin flushed, hot, dry, and scarlet; ataxia, restlessness, and excitement; hallucinations and delirium; coma

The clinical picture of a high (toxic) dose of atropine may be remembered by an old mnemonic device that summarizes the symptoms: *Red as a beet, Dry as a bone, Blind as a bat, Hot as firestone, and Mad as a hatter.*

Dose dependent effects of Atropine

Indications of Atropine

บัญชียาหลักแห่งชาติ ปี พ.ศ. 2567

2.3. Anti-arrhythmic drugs

1	Adenosine		sterile sol	ก
2	Atropine sulfate		sterile sol	ก

11.3. Mydriatics and cycloplegics

1	Atropine sulfate		eye drop	ค
2	Cyclopentolate hydrochloride		eye drop	ค
3	Phenylephrine hydrochloride		eye drop	ค
4	Tropicamide		eye drop	ค

15.1.5. Anticholinesterases and antimuscarinic drugs used in anesthesia

1	Atropine sulfate		sterile sol	ก
2	Neostigmine methylsulfate		sterile sol	ก
3	Edrophonium chloride ยากำพรั้		sterile sol (เฉพาะ 10 mg/mL)	ง
4	Glycopyrronium bromide		sterile sol	ง
5	Sugammadex		sterile sol (เฉพาะ 100 mg/mL) (2 mL)	ง

16.3 Pesticide & herbicide poisoning

1	Atropine sulfate		sterile sol	ก
2	Pralidoxime chloride	2-PAM	sterile pwdr	ก
3	Cyclophosphamide		sterile pwdr	ค

แบบฝึกหัดทบทวน

- 1. **Atropine works by blocking which type of receptors?**
 - A) Nicotinic receptors
 - B) Alpha-adrenergic receptors
 - C) Beta-adrenergic receptors
 - D) Muscarinic receptors
- 2. **Which of the following is a primary therapeutic use of atropine?**
 - A) Treatment of bradycardia
 - B) Management of hypertension
 - C) Treatment of glaucoma
 - D) Induction of miosis
- 3. **In cases of organophosphate poisoning, why is atropine administered?**
 - A) To increase acetylcholine levels
 - B) To block excessive acetylcholine activity
 - C) To inhibit acetylcholinesterase
 - D) To activate muscarinic receptors
- 4. **Which of the following conditions is a contraindication for the use of atropine?**
 - A) Bradycardia
 - B) Myasthenia gravis
 - C) Peptic ulcer disease
 - D) Parkinson's disease

Clinical uses of muscarinic antagonists

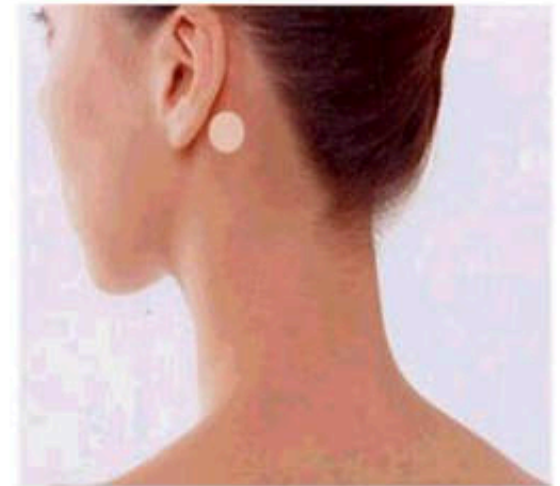
Systems	Indications	Drugs
Cardiovascular	Treatment of sinus bradycardia	Atropine
Ophthalmic	To dilate the pupil	Atropine , tropicamide or cyclopentolate eye drops
Neurological	Prevention of motion sickness	hyoscine (orally or transdermally) , scopolamine
	Parkinsonism	benzhexol, bengtropine
Respiratory	Asthma and chronic obstructive pulmonary disease	ipratropium or tiotropium (inhalation)
Anesthetic premedication	To dry secretions	atropine, hyoscine
Gastrointestinal	To facilitate endoscopy and gastrointestinal radiology by relaxing gastrointestinal smooth muscle	hyoscine
	antispasmodic in irritable bowel syndrome	dicycloverine (dicyclomine)
URINARY	Reduces detrusor smooth muscle tone, spasms	Oxybutynin

scopolamine



Hyoscine N-butylobromide

Ipratropium



Atropine



Muscarinic receptor antagonists for treatment of overactive urinary bladder

NONPROPRIETARY NAME	TRADE NAME	$t_{1/2}$ (HOURS)	PREPARATIONS ^a	DAILY DOSE (ADULT)
Oxybutynin	DITROPAN, others OXYTROL GELNIQUE	2-5	IR ER Transdermal patch Topical gel	10-20 mg ^b 5-30 mg ^b 3.9 mg 100 mg
Tolterodine	DETROL	2-9.6 ^c 6.9-18 ^c	IR ER	2-4 mg ^{b,d} 4 mg ^{b,d}
Trospium chloride	SANCTURA	20 35	IR ER	20-40 mg ^e 60 mg ^e
Solifenacin	VESICARE	55	IR	5-10 mg ^b
Darifenacin	ENABLEX	13-19	ER	7.5-15 mg ^f
Fesoterodine	TOVIAZ	7	ER	4-8 mg

^aPreparations are designated as follows: IR, immediate-release tablet; ER, extended-release tablet or capsule.

^bDoses may need to be reduced in patients taking drugs that inhibit CYP3A4.

^cLonger times in indicated ranges are seen in poor metabolizers.

^dDoses should be reduced in patients with significant renal or hepatic impairment.

^eDoses should be reduced in patients with significant renal impairment; dosage adjustments also may be needed in patients with hepatic impairment.

^fDoses may need to be reduced in patients taking drugs that inhibit CYPs 3A4 or 2D6.

1.2. Antispasmodics and other drugs altering gut motility

1	Dicycloverine hydrochloride	Dicyclomine hydrochloride	tab	ก
2	Domperidone		tab (as base/maleate), susp (as base/maleate)	ก
3	Hyoscine butylbromide	Hyoscine-n-butylbromide	tab, syr, sterile sol	ก
4	Metoclopramide		tab, syr, sterile sol	ก
5	Mebeverine hydrochloride		tab	ข

3.1.2. Compound antimuscarinic bronchodilators

1	Ipratropium bromide + Fenoterol hydrobromide		MDI, sol for nebulizer	ก
2	Tiotropium bromide		DPI (เฉพาะ 18 mcg)	ค

7.4.2. Drugs for urinary frequency enuresis, and incontinence

1	Oxybutynin hydrochloride		immediate release tab	ค
2	Trospium chloride		immediate release tab (เฉพาะ 20 mg)	ค

Oxybutynin hydrochloride immediate release tab

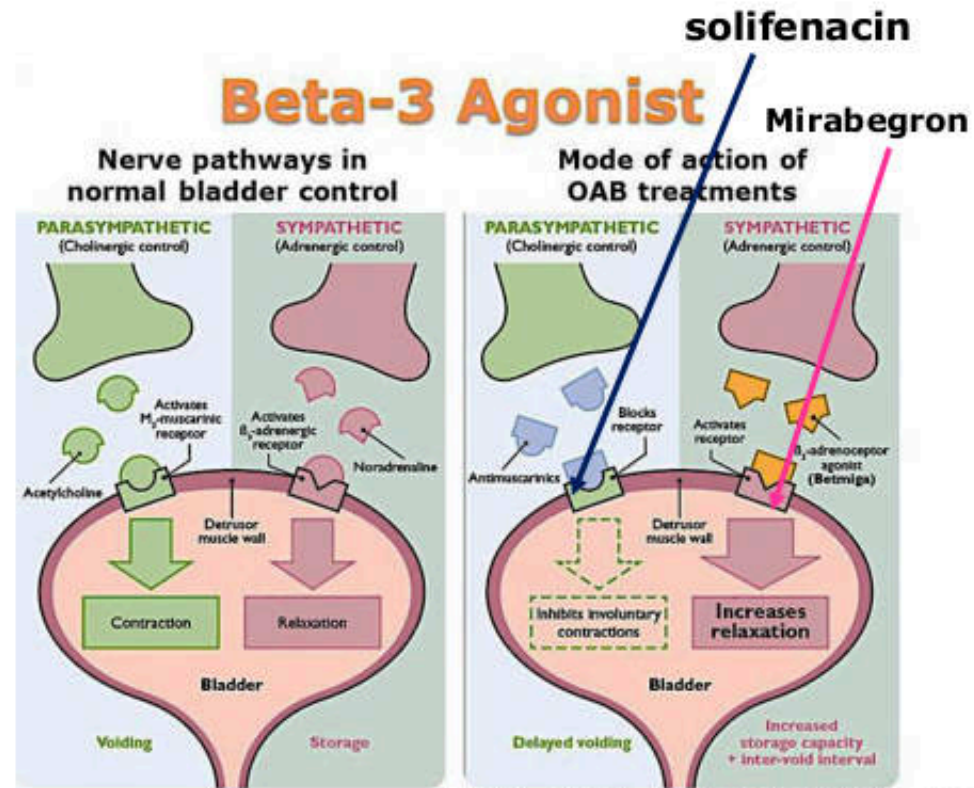
ค 7.4.2. Drugs for urinary frequency enuresis, and incontinence

เงื่อนไข :

1. ใช้สำหรับ neurogenic bladder ในผู้ป่วยเด็กตั้งแต่แรกเกิด (newborn)
2. ใช้ใน overactive bladder (OAB) ในผู้ป่วยเด็กตั้งแต่ 5 ปีขึ้นไป
3. ไม่ใช้สำหรับ nocturnal enuresis (ปัสสาวะรดที่นอนในเด็ก)

Solifenacin + Mirabegron for treatment of OAB

- US-FDA (2018): approved combination of **Mirabegron** and **solifenacin** for treatment of OAB
- **Expression of β_3 receptor:** brown adipose tissue, gallbladder, and ileum, white adipose tissue, and the **detrusor muscle of the bladder**
- **Mirabegron:** β_3 adrenergic receptor agonist approved by US-FDA (2012) for treatment of overactive bladder (OAB) and urinary incontinence
- Moderate CYP2D6 inhibitor
- **Side effects:** hypertension, increased incidence of urinary tract infection, headache



Adapted from Chu FM, Dmochowski R. Am J Med 2006;119(3 Suppl 1):3-8.

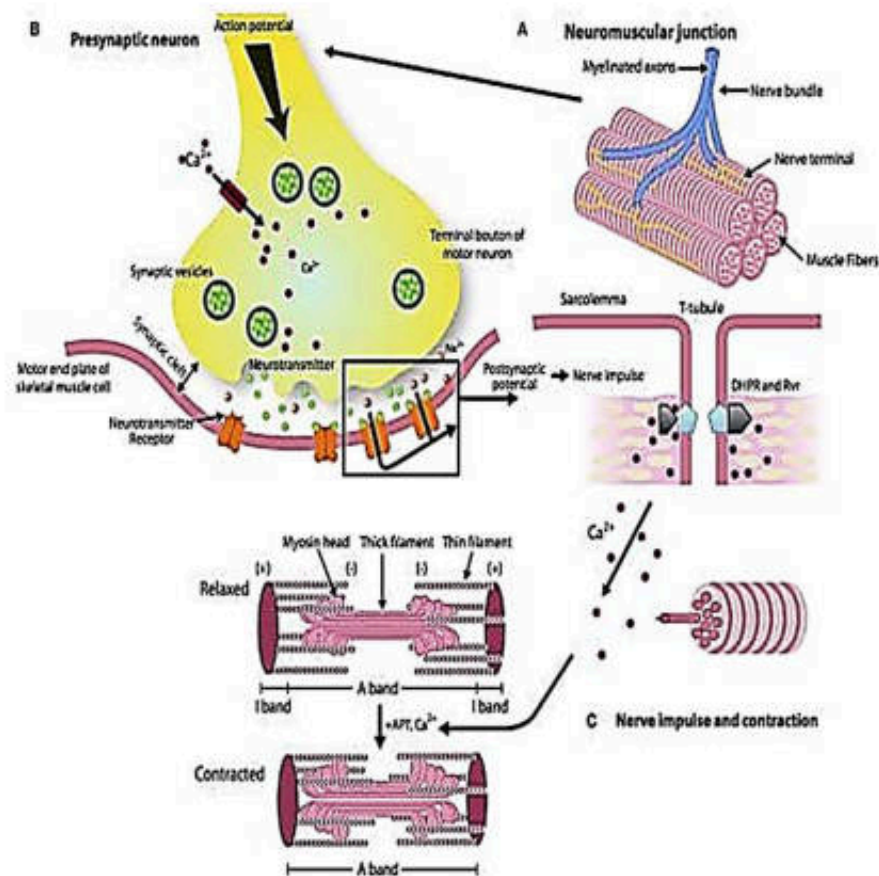
แบบฝึกหัด

ทบทวน

- 1. **Which of the following is a common side effect of anticholinergic drugs?**
 - A) Diarrhea
 - B) Excessive sweating
 - C) Dry mouth
 - D) Miosis
- 2. **Which of the following symptoms might indicate anticholinergic toxicity?**
 - A) Hypotension
 - B) Excessive salivation
 - C) Blurred vision and dry skin
 - D) Bradycardia
- 3. **Which anticholinergic agent is used in the management of Parkinson's disease?**
 - A) Atropine
 - B) Ipratropium
 - C) Benztropine
 - D) Oxybutynin
- 4. **What is a potential adverse effect of using anticholinergic drugs in elderly patients?**
 - A) Seizures
 - B) Increased cognitive function
 - C) Delirium and confusion
 - D) Hyperhidrosis

3. Neuromuscular blocking agents (NMB)

- Inhibit the binding of Ach and nicotinic receptor (N_m receptor) at Neuromuscular junction
- Prototype: Curare
- Adjunct to anaesthesia
- **Mechanisms:**
 - acting at presynaptic to inhibit ACh synthesis or release
 - Acting at postsynaptic
- **Clinical uses:**
 - Adjunct to anesthesia to relax skeletal muscle
 - Facilitate intubation



1. Non-depolarized blocking agents (competitive agents)

- **Mechanism of action:**
competitively block the binding of ACh receptor at the end plate
- **PKs:** Only intravenous forms
- **Duration of action:** varies between about 15 min - 2 h

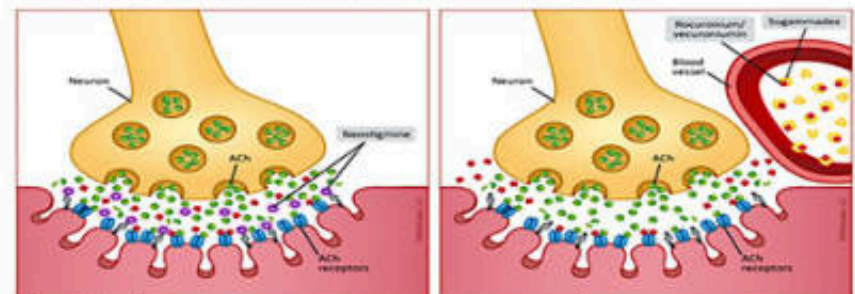
Drug	Speed of onset	Duration of action	Main side effects	Notes
Tubocurarine	Slow (>5 min)	Long (1–2 h)	Hypotension (ganglion block plus histamine release) Bronchoconstriction (histamine release)	Plant alkaloid, now rarely used Alcuronium is a semisynthetic derivative with similar properties but fewer side effects
Pancuronium	Intermediate (2–3 min)	Long (1–2 h)	Slight tachycardia Hypertension	The first steroid-based compound Better side effect profile than tubocurarine Widely used Pipcuronium is similar
Vecuronium	Intermediate	Intermediate (30–40 min)	Few side effects	Widely used Occasionally causes prolonged paralysis, probably owing to active metabolite Rocuronium is similar, with faster onset
Atracurium	Intermediate	Intermediate (<30 min)	Transient hypotension (histamine release)	Unusual mechanism of elimination (spontaneous non-enzymic chemical degradation in plasma); degradation slowed by acidosis Widely used Doxacurium is chemically similar but stable in plasma, giving it long duration of action Cisatracurium is the pure active isomeric constituent of atracurium, more potent but with less histamine release
Mivacurium	Fast (~2 min)	Short (~15 min)	Transient hypotension (histamine release)	Chemically similar to atracurium but rapidly inactivated by plasma cholinesterase (therefore longer acting in patients with liver disease or with genetic cholinesterase deficiency [see p. 172 and Ch. 11])

Reverse the action of non-depolarizing drugs

- Neostigmine
- Atropine (prevent unwanted parasympathomimetic effects)
- **Sugammadex**, a synthetic cyclodextrin
 - blockade effect of **rocuronium/ vecuronium**
 - **First selective relaxant binding agent for neuromuscular block reversal**
 - selectively binds neuromuscular blocking drugs →inactive complex in the plasma →excreted unchanged in the urine
 - produce more rapid reversal of block with fewer unwanted effects than neostigmine
 - regardless of the depth of the block

Neuromuscular block reversal

Neostigmine vs Sugammadex



Neostigmine

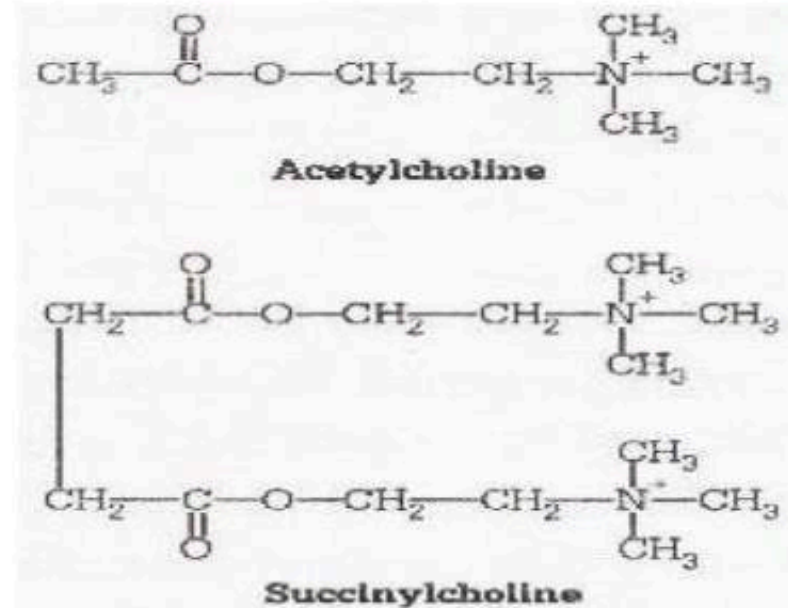
- Binds with any NMBA at the NMJ
- Has a ceiling effect and therefore cannot reverse deep NMB

Sugammadex

- Encapsulates rocuronium or vecuronium in the blood stream
- Rapidly reverses any level of NMB

2. Depolarizing blocking drugs

- “Suxamethonium” (Succinylcholine: SCh)
- Ach-like structure (2 ACh molecules linked by their acetyl groups)
- acts similarly to Ach but action lasts only a few minutes
- Quickly hydrolyzed by plasma cholinesterase
- cause depolarization block
- short acting and used during induction of anesthesia and intubation of the airway



Drug	Speed of onset	Duration of action	Main side effects	Notes
Suxamethonium	Fast	Short (~10 min)	Bradycardia (muscarinic agonist effect) Cardiac dysrhythmias (increased plasma K ⁺ concentration – avoid in patients with burns or severe trauma) Raised intraocular pressure (nicotinic agonist effect on extraocular muscles) Postoperative muscle pain	Acts by depolarisation of endplate (nicotinic agonist effect) – the only drug of this type still in use Paralysis is preceded by transient muscle fasciculations Short duration of action owing to hydrolysis by plasma cholinesterase (prolonged action in patients with liver disease or genetic deficiency of plasma cholinesterase) Used for brief procedures (e.g. tracheal intubation, electroconvulsive shock therapy) Rocuronium has similar speed of onset and recovery, with fewer unwanted effects

- ***No specific antagonistic agent for treating of toxicity***

Comparison of competitive and depolarized blocking agents

	D-TUBOCURARINE	DECAMETHONIUM
Effect of D-tubocurarine administered previously	Additive	Antagonistic
Effect of decamethonium administered previously	No effect, or antagonistic	Some tachyphylaxis; but may be additive
Effect of anticholinesterase agents on block	Reversal of block	No reversal
Effect on motor end plate	Elevated threshold to acetylcholine; no depolarization	Partial, persisting depolarization
Initial excitatory effect on striated muscle	None	Transient fasciculations
Character of muscle response to indirect tetanic stimulation during <i>partial</i> block	Poorly sustained contraction	Well-sustained contraction

Drug interactions

Drugs	Example of drugs	Interaction effect
General anesthetics	isoflurane, enflurane	synergism
Antibiotic i.e. Aminoglycoside	gentamicin, kanamycin, amikacin	synergism
Local anesthetic	lidocaine	synergism
Anticholinesterases	Neostigmine	Antagonism
Calcium channel blocker	amlodipine, felodipine, nifedipine, diltiazem	synergism
Non-depolarizing blocking agents	Vecuronium, Mivacurium	antagonism

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15.1.3. Muscle relaxants (Neuromuscular blocking drugs)				
1	Atracurium besilate		sterile sol	ค
2	Cisatracurium besilate		sterile sol	ค
3	Pancuronium bromide		sterile sol	ค
4	Rocuronium bromide		sterile sol	ค
5	Suxamethonium chloride	Succinylcholine chloride	sterile pwdr, sterile sol	ค
6	Vecuronium bromide		sterile pwdr	ค
7	Dexmedetomidine		sterile sol	ง

Dexmedetomidine

sterile sol

(เฉพาะ 100 mcg/ml) (2 ml)

ง

15.1.4 Sedative and analgesic peri-operative drugs

เงื่อนไข :

1. สำหรับการสงบประสาทและระงับปวดในหออภิบาลผู้ป่วยวิกฤต (for sedation and analgesia in Intensive Care Unit patients) เฉพาะกรณีผู้ป่วยอายุมากกว่า 65 ปี หรือ มีความเสี่ยงสูงต่อ delirium โดยโรคประจำตัว หรือมีข้อห้ามใช้ยาอื่นในบัญชียาหลักแห่งชาติ
2. สำหรับการสงบประสาทและระงับปวดสำหรับหัตถการทางการแพทย์ (for procedural sedation and analgesia)
3. ใช้ร่วมกับการระงับความรู้สึก ในระหว่างผ่าตัด (for perioperative adjuvant anesthetic management)

แบบฝึกหัดทบทวนบทเรียนเรื่อง Parasympatholytic drugs (Anticholinergic agents)

1. จงอธิบายประโยชน์และอาการไม่พึงประสงค์ของ **atropine**
2. จงอธิบายว่าเหตุใดยา **atropine** จึงใช้เป็นหนึ่งในส่วนประกอบของยาชนิด **diphenoxylate** ที่ใช้เป็นยาหยุดถ่ายรักษาอาการท้องเสีย
3. จงอธิบายความแตกต่างระหว่างยา **oxybutynin** กับ **darifenacin** ในการรักษาอาการ **overactive bladder**
4. เหตุใดการใช้ยากลุ่ม **NMJ blocker** กลุ่ม **non-depolarized block** ร่วมกับ **depolarized block** จึงเกิดอันตรกิริยาระหว่างกัน
5. อธิบายกลไกการออกฤทธิ์ และความแตกต่างระหว่าง **Sugammadex** กับยา **Neostigmine** ในการต้านพิษจากยา **vecuronium**